

Is Your Full Production Model Less Accurate Than My Local Pruned Model? Comparing LLM Compression and Production Noise Floors

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Abstract

Large Language Models (LLMs) are in many cases compressed for local or user deployment to minimize the VRAM needed and the execution latency. Because compression induces a performance loss, it is assumed that cloud production environments serving full-precision models represent the quality baseline. In this work, we challenge this assumption by exposing a key difference: cloud production systems process queries via dynamic batching ($B \geq 32$), where hardware-level floating-point non-associativity introduces reduction jitter and logit variance. In contrast, local single-user deployments typically execute deterministically at batch size $B = 1$. We argue that benchmarking locally compressed models against unbatched ($B = 1$) full-precision baseline evaluates compression against a theoretical baseline, not against an operational one. By comparing the compressed model to the real-world dynamic batching noise floor, we show that single-user execution yields a determinism surplus. This operational slack provides a bounded error budget, allowing local models to be pruned without dropping below the output fidelity of full-precision cloud endpoints.

1 Quick test: how much can we gain

We can use this library:

<https://github.com/NVIDIA/Model-Optimizer>

with different pruning levels and measure range on the logprobs vs uncompressed and compare with the $B = 1$ results we have from the other paper for a couple of models to find the percentage of pruning that gives the same range.

It seems the code is simple?

Configure structural pruning target (e.g., `prune 15`) `prune_config = "pruning_mode" : "mha_and MLP", "export format"`

Apply ModelOpt structural pruning `model_pruned = mtp.prune(model_pruned, config = prune_config, forward_calibration_loop,)`